MAT 3007 – Optimization

Spring Semester, CUHK-Shenzhen

Instructor: Junchi Yang

Course Description

This course introduces the foundational concepts and practical applications of optimization. The first part of the course focuses on *linear programming*, covering problem formulation, the simplex method, and duality theory. In the second part, the course shifts to *nonlinear optimization*, where we explore optimality conditions and key algorithms. The course concludes with an introduction to *integer programming*, including basic algorithms and applications.

Lecture Location and Time

- Location: TxC301
- Lecture Session 01: Monday, 3:30 p.m. – 4:50 p.m.
- Lecture Session 02: Wednesday, 3:30 p.m. – 4:50 p.m.

Instructor

- Junchi Yang
- Email: `yangjunchi@cuhk.edu.cn`
- Office: Dao Yuan 321a
- Office Hours: Wednesday, 11:00 a.m. – 12:00 p.m.

Teaching Assistants

- **Wenjun Zheng**
  Email: `118010446@link.cuhk.edu.cn`
  Office: CD (Cheng Dao) 318B
  Office Hours: Friday, 8:00 – 9:00 p.m.
- **Zhuohan Wang**
  Email: `224040339@link.cuhk.edu.cn`
  Office: RB (Research Building B) 302
  Office Hours: Wednesday, 8:00 – 9:00 p.m.
- **Jiaming Chen**
  Email: `121090033@link.cuhk.edu.cn`

Tutorial Times

- T-1: Monday, 7:00 – 7:50 p.m., Teaching C Building 414
- T-2: Monday, 8:00 – 8:50 p.m., Teaching C Building 414
- T-3: Wednesday, 6:00 – 6:50 p.m., Teaching C Building 414
- T-4: Wednesday, 7:00 – 7:50 p.m., Teaching A Building 302
- T-5: Thursday, 6:00 – 6:50 p.m., Teaching C Building 308

Note: Tutorials will begin in the **second week** of the semester.

Reference

Slides and other supplementary materials will be provided after each lecture.

Reference Books:

- Bertsimas, D., Tsitsiklis, J. *Introduction to Linear Optimization.* Athena Scientific. (1997).
- Boyd, S., Vandenberghe, L. *Convex Optimization.* Cambridge University Press. (2004).
- Beck, A.*Introduction to Nonlinear Optimization: Theory, Algorithms, and Applications with MATLAB.* MOS-SIAM Series Optimization. (2014).

Grading Scheme

- **Assignments:** 20%
  There will be 9 assignments this semester, and the lowest score will be dropped. Late submissions are not accepted.
  Submission Guidelines: Plagiarism is strictly prohibited. Discussion with peers is encouraged, but each student must submit their own work. Submissions must be made electronically.
- **Midterm Exam:** 40%
  Schedule: Saturday, March 22nd, 10:00 a.m. – 11:30 a.m.
  Students may bring one self-prepared A4 cheat sheet (notes on both sides).
- **Final Exam:** 40%
  Schedule: To be announced.

Course Topics

The course is structured as follows:

- **Week 1:** Introduction to optimization.
- **Weeks 2 – 4:** Basics of linear optimization and the simplex method.

- **Weeks 4 – 5:** Duality theories for linear programs.
- **Weeks 6 – 7:** Dual simplex, sensitivity analysis, and advanced topics in linear programming.
- **Weeks 7 – 8:** Review and midterm exam.
- **Weeks 8 – 11:** Nonlinear optimization: Optimality conditions, convexity, and foundational concepts.
- **Weeks 11 – 13:** Algorithms for nonlinear problems.
- **Weeks 13 – 15:** Introduction to integer programming and basic algorithms.